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UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 1

2024/2025

SECP3723-01 SYSTEM DEVELOPMENT TECHNOLOGY

SECTION 01

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SYSTEM REQUIREMENT

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System requirement

1. Hardware Interfaces

The Koperasi KADA's Automated System will interface with standard hardware components, including desktop computers, laptops, and mobile devices. The software will support input devices such as keyboards and mice for desktops, as well as touchscreens for mobile devices, enabling users to choose their preferred method of interaction. Data and control interactions between the software and hardware will occur over standard web protocols, specifically HTTP and HTTPS. These protocols ensure secure and efficient data transmission to protect user information from unauthorised access and enhance system security.

User inputs, including clicks, taps, and keystrokes, will be processed by the software. In response, the system will trigger appropriate actions, such as displaying requested data, saving input to the database, or initiating processes like membership applications and loan requests. This design allows for seamless control and interaction across all devices which facilitates a smooth user experience that meets the needs of both staff and members.

2. Software Interfaces

This section describes the system's main software components, data flow and implementation constraints for the system. The architecture ensures efficiency and security while integrating a variety of technologies and components to enable smooth user-system interaction.

A seamless user experience and effective data management are provided by the system's architecture, which is made up of a number of crucial parts. Some of the main types of specific software components used in this project are database components of MySQL that are essential for storing such as membership records, loan applications and user information. Additionally, operating systems such as Windows, MacOS offer administrators and users a user-friendly interface. Next, core libraries such as HTML, PHP, JavaScript are used to create both the backend and the user interface. Moreover, Tools such as Bootstrap for responsive and Sublime Text for code management provide a framework for creating responsive web applications and integrated development environments for code management and writing. Finally, integrated commercial components, such as emails, are used for informing users and providing confirmation messages.

Additionally, for efficient administration and communication, data flow inside the system is essential. This allows a clear understanding of the incoming and outgoing data items' functions in the program's operation. The structure of incoming data includes several key elements. Firstly, user registration data is essential for setting up new user accounts, confirming identities and guaranteeing security by using unique accounts. In addition, login data plays a crucial role by establishing users' access rights and authenticating them to guarantee safe application access control. Furthermore, membership application data is collected and stored to manage membership requests and allows for tracking of each member's status. Moreover, loan application data is utilized by the system to process applications and determine loan eligibility, guaranteeing that users are provided with correct loan details.

Moreover, maintaining an organised structure requires cooperation between software components. The kinds of information shared by different components including

some of the data such as user data, application data, notifications and emails and the statement and report.

Lastly, specific implementation constraints must be followed in order to provide a secure and dependable application. This involves protecting user data across all components by utilising a centralised database, enforcing tight access rules and encrypting sensitive data like passwords.

3. Communications Interfaces

The Koperasi KADA Automated System requires a range of communication functions to support efficient data exchange, user notifications and secure interactions between system components and users. These interfaces will include support for email notifications, web browser accessibility, and electronic form submissions to streamline user interactions and enable timely updates.

3.1. Email Communication

The system will use Simple Mail Transfer Protocol (SMTP) to enable automated email notifications. Messages sent to users will be formatted with clear and concise content, using plain text and HTML for better compatibility with different email clients. Email notifications will also include a standardized header to specify the source and purpose of each message, ensuring consistency across all communications. For security, email content will avoid including sensitive information.

3.2. Web Browser Accessibility

The system must be accessible through standard web browsers, including Google Chrome, Mozilla Firefox and others ensuring broad compatibility for users with different browser preferences. HTTP/HTTPS protocols will be employed to facilitate secure browser-server communication, with HTTPS enforced for all transactions involving sensitive data. The use of HTTPS with SSL/TLS encryption will help protect data from interception during transit, ensuring data integrity and privacy for users accessing the system over the internet.

3.3. Electronic Form Submissions

The system will incorporate electronic forms for key functions such as loan applications, membership registrations and information updates. These forms will adhere to the HTML5 standard, ensuring they are compatible across devices and accessible on various platforms. Form data will be validated both client-side and server-side to prevent entries and enhance security. Sensitive fields in these forms, such as financial data, will be encrypted before being stored in a database to safeguard user information.

3.4. Communication Security and Encryption

Given the sensitivity of the data handled by the system, security protocols will be strictly enforced. SSL/TLS encryption will be implemented for all HTTP communications and sensitive data exchanged over the network will be encrypted to prevent unauthorized access. Additionally, secure access control measures will be integrated, including user authentication and role-based authorization to restrict access to different system components.

4. Performance Requirements

The system should handle up to 500 concurrent users without performance degradation to ensure timely processing of membership and loan applications. The response time for any user request should not exceed 2 seconds under normal operating conditions. For real-time operations, such as automated notifications and data retrieval, the latency should be kept below 1 second. In addition, optimal performance should be maintained even during peak load times so that all users receive a fast service. Regular performance monitoring and optimization is performed to proactively address any potential issues. These requirements aim to provide an efficient user experience which is critical to the effectiveness of the system.

5. Safety Requirements

The system includes specific safety requirements to mitigate potential loss, damage, or harm associated with its use, particularly in handling sensitive financial and personal data. A primary safety concern is the safeguarding of data integrity and confidentiality. The system will incorporate robust encryption methods, such as SSL/TLS, for all data transmissions to prevent interception by unauthorized parties. Additionally, data stored on the server, including member and loan information, will be encrypted to provide another layer of protection against unauthorized access.

The system will also include safety mechanisms to prevent accidental or unauthorized changes to critical data, such as loan applications and membership records. Role-based access control will be implemented, allowing only authorized personnel to make updates or approve applications, which minimizes the risk of accidental or malicious alterations.

Another important safety aspect is the prevention of service interruptions that could disrupt the organization's operations. Regular backups will be scheduled to create copies of critical data, ensuring that information can be recovered in case of accidental deletion, hardware failure or security incidents.

Finally, the system's design will comply with external safety and data protection regulations, particularly those related to financial data handling. Compliance with regulations will include securing user consent for data collection, ensuring data accuracy, and allowing users to update or delete their data where required. These measures align the system with industry standards for data security, reducing legal risks and promoting user confidence.

6. Security Requirements

For this system, security and privacy are paramount. All user activities, including logins, data access, and changes to user permissions must be logged for accountability. The system will require multi-factor authentication (MFA) to help prevent unauthorized access. Additionally, data handling in the system will follow basic privacy guidelines, like those outlined in GDPR, to help protect user information. User data will be protected by basic encryption standards, and sensitive actions, such as password changes, will have

additional security checks. The system will also include periodic reviews to help identify any new security risks or improvements. Finally, user access levels will be clearly defined to limit exposure to sensitive data based on roles.

7. Software Quality Attributes

The Software Quality Attributes section outlines crucial features that support the system's efficacy, efficiency and user satisfaction. These attributes lay the foundation for long-term success and user-friendliness by ensuring that the product satisfies both technical and user criteria. By defining essential attributes such as maintainability, portability, reliability, usability and availability, we label core needs for platform accessibility, future updates and consistent performance. The next subsections go into great detail about the importance of these features and how the system is intended to use them.

7.1. Maintainability

The system required to be built with simple bug fixes and updates in mind, without compromising other features. This contains a modular design that guarantees little inconvenience during maintenance procedures which enables independent component modifications or replacement. Maintainability should be measurable by laying objectives to finish major updates in a week and minor updates in 1 to 2 days.

7.2. Portability

The software needs to work with a variety of operating systems. By doing this, the system works in a variety of hardware and software environments, including different Windows and macOS versions. Testing portability involves deploying the program in different surroundings and making sure that performance remains consistent.

7.3. Reliability

The system must secure data consistency and correctness, with a monthly downtime tolerance of no more than 0.5%. The implementation of automated error-handling systems and frequent data backups is necessary to defend against

data loss and ensure a smooth recovery in case of failure.

7.4. Usability

An user-friendly interface should enable new users to complete tasks without requiring a lot of training. The objective of usability testing should be a high user satisfaction rate and an average task completion time that falls within acceptable limits.

7.5. Availability

The system should be available for users to use whenever they need. This is especially crucial for systems that facilitate real-time activities. Availability should remain accessible and functional even under interruptions or heavy loads. A target of 99% available rate can be set as a benchmark to help on the availability testing.